

KONDA MADANA GOPALA REDDY

Mechatronics & Mechanical Engineer | MATLAB · Simulink · Python · Control Systems | Open to Werk student & Full-Time Roles in Germany

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PROFESSIONAL SUMMARY

MSc Mechatronics & Mechanical Engineering candidate (Hochschule Kaiserslautern, 2025–present) with a B.Tech in Mechanical Engineering and proven hands-on experience designing closed-loop control systems, dynamic system models, and mechatronic prototypes in MATLAB/Simulink. Proficient in nonlinear state-space modelling, numerical ODE/PDE solvers, and sensor-actuator integration. Seeking Werkstudent, internship, or entry-level R&D roles in control systems, predictive maintenance, or mechatronic product development in Germany.

EDUCATION

M.Sc. Mechatronics and Mechanical Engineering | Hochschule Kaiserslautern, Germany

March 2025 – Present

B.Tech Mechanical Engineering | JNTU Kakinada, India

January 2020 – July 2024

INTERNSHIP EXPERIENCE

Production Intern | ZF Commercial Vehicle Solutions — Exhaust Valve Production

2024

- Supported shop-floor production processes for exhaust valve manufacturing.

Production Intern | Visakhapatnam Steel Plant — Roll Production for Steel Rod Manufacturing

2023

- Supported production of rolls used in steel rod manufacturing; authored internal technical documentation on the process.

APPLIED CONTROL & MECHATRONIC SYSTEMS PROJECTS

Inverted Pendulum Control & Two-Wheel Robot Drive Optimisation · MATLAB, Simscape Multibody, LQR, Stateflow

- Designed an LQR (Linear-Quadratic Regulator) full-state feedback controller in MATLAB and Simscape Multibody for an inherently unstable self-balancing two-wheel robot, validated through linearization and Model Linearizer gain tuning.
- Layered Stateflow-based phase logic (pre-acceleration, sprint, brake) on top of a position–velocity feedback scheme to drive the physical robot as fast as possible while holding balance.
- Achieved a 3 m traverse in 2.56–2.59 s across repeated hardware trials with zero balance loss (avg. 1.07–1.10 m/s).
- Extended the controller to driving-with-turns, weight-compensation (cascaded PID recovering in 5.0–7.3 s from a 650 g off-center load), and figure-8 tracking challenges.

Gimbal-Based 3D Printer Printhead – Mechatronic Prototype · Siemens NX, Altair Inspire, Altair MotionView

- Designed a multi-axis gimbal-driven printhead in Siemens NX to eliminate support-material dependency in FDM printing, where conventional setups can waste up to 40% material on overhangs and undercuts.
- Modelled a 5-DOF rotational kinematic chain and ran topology optimization in Altair Inspire to lighten and stiffen the frame.
- Validated joint torque loading and trajectory generation through multi-body simulation in Altair MotionView, confirming both drive motors were correctly sized with torque headroom to spare.
- Achieved 5 resultant DOF at the nozzle (3 translational, 2 rotational) and validated both a smooth curve-based staircase toolpath and a spiral vase toolpath.

Rigid Pendulum with Damping – Numerical Solver Benchmarking · MATLAB, Midpoint Method, ode45, Numerical Methods

- Implemented a Midpoint method (2nd-order Runge-Kutta) solver from scratch in MATLAB and benchmarked it against MATLAB's adaptive ode45 solver for a damped nonlinear pendulum.
- Swept step count N from 500–2500 to quantify convergence, and visualised phase portraits, vector fields, and mechanical energy decay to validate solver behaviour.

- Extended the model to a real-world case study — an ABB IRB120 robot arm (Links 3–6) modelled as an equivalent pendulum ($L = 0.446$ m, derived from real mass/inertia properties).
- Found the Midpoint method tracked ode45 to within 10^{-4} – 10^{-3} rad across all tested resolutions, converging to a stable error floor beyond $N \approx 2000$ — informing solver choice for subsequent dynamic modelling work.

Vehicle Cruise Control – Closed-Loop Longitudinal Dynamics Simulation · MATLAB/Simulink, Feedback Control, Model-in-Loop (MiL)

- Built a physics-based vehicle plant model in MATLAB/Simulink, capturing drag and rolling resistance dynamics under variable road gradient.
- Designed and tuned a PI feedback controller via model-in-loop (MiL) simulation to maintain target velocity without actuator saturation.
- Ran disturbance rejection scenarios for $\pm 10\%$ grade change across 5 road gradient conditions.
- Achieved $< 2\%$ steady-state velocity error and full disturbance rejection within a 3 s settling time.

TECHNICAL COMPETENCIES

Programming & Simulation: MATLAB · Simulink (Model-in-Loop) · Python · Numerical ODE/PDE Solvers · State-Space Modelling · Signal Processing

Control Systems: LQR · PID Controller Design · Full-State Feedback · Stateflow · Nonlinear Dynamics · Closed-Loop System Tuning

Engineering Design & CAD: Siemens NX · CATIA V5 · AutoCAD · Altair Inspire · Altair MotionView · Kinematic Chain Modelling · Mechatronic System Design

Mechanical Systems: CNC Machine Operation & Maintenance · Preventive Maintenance · Fault Diagnosis · Tolerance Analysis

Interests / Target Domains: Predictive Maintenance · Sensor Data Analytics · Pattern Recognition · AI-Driven Industrial Tools · R&D Environments

LANGUAGES

English – Fluent (C1) | German – A2 (actively improving) | Telugu – Native

SOFT SKILLS & ATTRIBUTES

Cross-functional Teamwork · Technical Communication · Time Management · Adaptability · Self-driven Learner